

The New York City Department of Education

2023 Specialized High Schools Admissions Test

GENERAL DIRECTIONS

Student Name: _____

Identifying Information

Turn to Side 1 of the answer sheet.

Notify the proctor immediately if you are ill or should not be taking this test. Do not sign the statement or begin the test. Return your answer sheet to the proctor.

Line 1: Read the statement and sign your name in the space following the word “signature.” Do not print your name.

Line 2: Print today’s date, using the numbers of the month, the day, and the year. For example, the date September 21, 2023, would be 9-21-23.

Line 3: Print your birth date, using the numbers of the month, the day, and the year. For example, a birth date of March 1, 2006, would be 3-1-06.

Grid 4: Print the letters of your first name, or as many as will fit, in the boxes. If you have a middle initial, print it in the box labeled “MI.” Then print the letters of your last name, or as much as will fit, in the boxes provided. Below each box, fill in the circle that contains the same letter as the box. If there is a space or a hyphen in your name, fill in the circle under the appropriate blank or hyphen.

Make dark marks that completely fill the circles. If you change a mark, be sure to erase the first mark completely.

Grid 5:

1. Print the name of the school where you are now enrolled in the space at the top of the grid.
2. In the boxes marked “SCHOOL CODE,” print the six-digit code that identifies your school. Below each box, fill in the circle that contains the same letter or number as the box. (You can find your school code on your Test Ticket. If it is not there, tell the proctor, and the proctor will get the school code for you.)
3. If you attend a private or parochial school, fill in the circle marked “P.”

Grid 6: Complete the grid with your date of birth. Print the first three letters of the month in the first box, the number of the day in the next box, and the year in the last box. Then fill in the corresponding circles.

Grid 7: Print your student ID number in Grid 7. You can find your student ID number on your Test Ticket. In the boxes, print your nine-digit student ID number. Below each box, fill in the circle containing the same number as in the box.

Grid 8: In most cases, Grid 8 is already filled in for you. If it is not, copy the letter and numbers shown in the upper-right corner of your test booklet into the boxes. Below each box, fill in the circle containing the same letter or number as the box.

Now review Side 1 to make sure you have completed all lines and grids correctly. Review each column to see that the filled-in circles correspond to the letters or numbers in the boxes above them.

Turn your answer sheet to Side 2 and flip this booklet over to continue with General Directions.

PART 1 — ENGLISH LANGUAGE ARTS

57 QUESTIONS

REVISING/EDITING

QUESTIONS 1-9 (PART A AND PART B)

REVISING/EDITING PART A

DIRECTIONS: Read and answer the following questions. You will be asked to recognize and correct errors so that the sentences or short paragraphs follow the conventions of standard written English. You may write in your test booklet as needed to take notes. You should reread relevant parts of the sentences or paragraphs, while being mindful of time, before marking the **best** answer for each question.

1. Which revision should be made to correct the sentence?

To surprise her dad for his birthday on Friday, Yolanda ordered tempting savory sandwiches from a deli in her neighborhood, prepared his favorite chocolate cake, and invited his best friends over for the evening.

- A. Delete the comma after **Friday**.
- B. Insert a comma after **tempting**.
- C. Insert a comma after **deli**.
- D. Delete the comma after **neighborhood**.

2. Which sentence in the paragraph contains an error in its construction and should be revised?

(1) On the evening of July 13, 2019, a major power outage affected the Upper West Side of Manhattan in New York City. (2) Leaving approximately 73,000 residents without electricity for three long hours, lights did not function, refrigerators did not stay cold, and air conditioners did not work. (3) Longtime city residents were particularly confused because an eerily similar event had occurred years earlier—on the exact same day! (4) The famous New York City Blackout of 1977, which lasted for 25 hours, also happened on July 13, an odd coincidence to say the least.

- E. sentence 1
- F. sentence 2
- G. sentence 3
- H. sentence 4

3. Which pair of revisions is needed to correct the errors in the paragraph?

(1) Yalina, Michael, and Malcolm love making pancakes with their granddad on Saturday mornings. (2) Yalina's job is to open the box and pour the pancake mix into a bowl, slowly adding water, eggs, melted butter, and blueberries. (3) Michael uses a wooden spoon to vigorously stir the mixture until it is smooth, and Malcolm helps Granddad carefully pour the batter onto a griddle one-fourth cup at a time. (4) Granddad turns each pancake when they start to bubble, while all three siblings get the table ready for a sweet delicious breakfast.

- A. Sentence 1: Delete the comma after **Yalina**, AND change **their** to **her**.
- B. Sentence 2: Change **is** to **are**, AND delete the comma after **bowl**.
- C. Sentence 3: Change **it is** to **they are**, AND delete the comma after **smooth**.
- D. Sentence 4: Change **they start** to **it starts**, AND insert a comma after **sweet**.

REVISING/EDITING PART B

DIRECTIONS: Read the text below and answer the questions following it. You will be asked to improve the writing quality of the text and to correct errors so that the text follows the conventions of standard written English. You should reread relevant parts of the text, while being mindful of time, before marking the **best** answer for each question.

The End of an Era

(1) In 2004, the National Aeronautics and Space Administration (NASA) landed a rover, or robotic vehicle, named *Opportunity* on the surface of Mars. (2) The rover's mission was to search for evidence of water and life on the planet. (3) The rover was designed to gather data on Mars for about 90 days, but *Opportunity's* assignment did not come to an end until nearly 15 years later in 2019.

(4) During the rover's mission, a high-resolution camera on the rover's robotic arm took magnified photographs of the surface of Mars. (5) The photographs revealed small round rocks scattered across the surface of the planet. (6) The scientists nicknamed these rocks "blueberries" based on the rocks' resemblance to the fruit. (7) These rocks were important because their spherical shape suggested that liquid water may have flowed over them for a substantial amount of time.

(8) *Opportunity* continued to provide valuable data about craters and other surface features.

(9) Throughout the mission, the rover traveled a total of 28.06 miles across Mars' surface.

(10) When *Opportunity* had traveled nearly the length of a marathon, scientists celebrated the accomplishment by naming the valley that the rover was in Marathon Valley. (11) However, in 2018, a dust storm spread all over the planet, putting the rover's progress on hold.

(12) Prior to this planet-wide dust storm, minor dust storms had periodically deposited a layer of dust onto the solar panels of the rover. (13) These prevented it from charging its batteries.

(14) Scientists then had to rely on the passing winds and dust devils to clean off the solar panels.

(15) After the massive 2018 dust storm subsided, scientists expected to be able to reestablish communication with *Opportunity*. (16) They made repeated attempts but to no avail. (17) In 2019, NASA finally declared *Opportunity's* mission complete.

(18) *Opportunity* was an important asset in scientists' study of Mars; it returned valuable data about our neighbor planet and survived powerful dust storms. (19) NASA described the mission of *Opportunity* as "one of the most successful and enduring feats of interplanetary exploration."

(20) Even though this rover's journey ended, other rovers still roam the surface of Mars.

4. Which sentence should follow sentence 3 to **best** introduce the topic of the passage?
- E. Despite the challenges of operating on the surface of another planet, the *Opportunity* rover made many valuable contributions to the field of space research.
 - F. The *Opportunity* rover mission was a success because it informed NASA scientists about how water might have existed on Mars.
 - G. Throughout the historic mission of the *Opportunity* rover, NASA scientists had to overcome challenges created by the flaws in the rover's design.
 - H. The *Opportunity* rover mission surpassed the original timeline of the assignment and allowed scientists to collect additional data about the planet Mars.
5. Which word should be added to the beginning of sentence 8 to provide a better transition to the third paragraph (sentences 8–11)?
- A. Overall,
 - B. Furthermore,
 - C. Meanwhile,
 - D. Consequently,
6. Which revision of sentence 11 **best** maintains the formal style established in the passage?
- E. However, in 2018, a dust storm covered the planet, bringing the rover's progress to a standstill.
 - F. However, in 2018, a dust storm shrouded the planet, impermanently halting the rover's progress.
 - G. However, in 2018, a dust storm enveloped the whole of the planet, arresting the rover's progress for a time.
 - H. However, in 2018, a dust storm covered up the whole planet, which quickly brought a stop to the progress of the rover.

7. What is the **best** way to combine sentences 12 and 13?
- A. Preventing the rover from charging its batteries, minor dust storms had periodically deposited a layer of dust onto the solar panels of the rover prior to this planet-wide dust storm.
 - B. Periodically depositing a layer of dust onto the solar panels of the rover prior to this planet-wide dust storm, minor dust storms had prevented it from charging its batteries.
 - C. Minor dust storms, prior to this planet-wide dust storm, prevented the rover from charging its batteries because they had periodically deposited a layer of dust onto the solar panels of the rover.
 - D. Prior to this planet-wide dust storm, minor dust storms had periodically deposited a layer of dust onto the solar panels of the rover, preventing it from charging its batteries.

8. Read this sentence.

One of *Opportunity's* major contributions was finding possible evidence of the presence of water on Mars.

Where should the sentence be added to the passage?

- E. between sentences 2 and 3
 - F. at the beginning of the second paragraph (before sentence 4)
 - G. at the end of the second paragraph (after sentence 7)
 - H. between sentences 8 and 9
9. Which concluding sentence should be added after sentence 20 to **best** support the information presented in the passage?
- A. These rovers, including one named *Curiosity*, will help scientists answer their lingering questions about the planet.
 - B. Today's rovers continue to add to *Opportunity's* discoveries and will help us learn even more about the planet.
 - C. And in conjunction with these rovers, NASA is using other resources, including an orbiting spacecraft known as *Mars Odyssey*, for the exploration of the planet.
 - D. With its red dust, rocky surface, and freezing temperatures, Mars will continue to be a challenging environment for today's rovers.

READING COMPREHENSION

QUESTIONS 10–57

DIRECTIONS: Read each of the following six texts and answer the related questions. You may write in your test booklet as needed to take notes. You should reread relevant parts of each text, while being mindful of time, before marking the **best** answer for each question. Base your answers only on the content within the text.

CONTINUE TO THE NEXT PAGE ►

Excerpt from “The Food Business Incubator”

by Mandalit Del Barco

- 1 “La Cocina” means “the kitchen” in Spanish. It’s also the name of a business incubator¹ based in San Francisco’s Mission District. Since it began in 2005, it’s been helping local food entrepreneurs, many of whom are low-income immigrant women, develop their small businesses.
- 2 Over the years, many of its alumni have found success: more than 50 chefs in its program have become self-sufficient business owners, and many of them have opened their own brick-and-mortar restaurants. Two alumnae of its culinary program . . . were even recognized as semi-finalists for prestigious James Beard awards.²
- 3 A new cookbook, *We Are La Cocina: Recipes in Pursuit of the American Dream*, tells some of their stories.
- 4 Executive director Caleb Zigas says the nonprofit La Cocina grew out of two grassroots economic development organizations³ who found many people cooking at their homes and selling food on the streets. The vendors needed an affordable commercial kitchen space and technical assistance in order for their businesses to be legally viable. La Cocina provided just such a space, in addition to helping them develop business plans, pull city permits and more.
- 5 Zigas says as many as eight businesses can work in the kitchen space at La Cocina at any one time. Some can prep for a farmers’ market sale, corporate catering gigs or weddings, while others might be making and packaging their food products.
- 6 “It’s just an incredible and exciting range of techniques, flavors, perspectives, age, language. And that’s a really beautiful thing,” he says. . . .
- 7 Twice a year, La Cocina hosts “F&B: Voices from the Kitchen,” a storytelling project where their chefs can tell their own stories, as they do in the new cookbook. Some of the chefs will be on tour to promote the book, whose proceeds will go to support La Cocina entrepreneurs.
- 8 NPR⁴ caught up with . . . La Cocina chefs, who shared their stories:

Mariko Grady, Aedan Fermented Foods

- 9 At La Cocina, you can often hear Mariko Grady singing or humming as she prepares miso, *koji*, and *amasake*. Her fermented products come in four different flavors, including mushroom and chicken, to be used in soups and sauces. She originally brought the fermenting rice and barley

¹**business incubator:** company that helps new and startup companies develop by providing services such as management training and office space

²**James Beard awards:** an award named after American cook and author James Beard, considered one of the highest honors in the culinary industry

³**economic development organizations:** organizations whose mission is to promote economic improvement within a specific geographical area

⁴**NPR:** National Public Radio, a nonprofit media organization that produces and distributes news and cultural programming

kōji seeds from Japan, where she had a 30-year career as a singer and dancer with the prestigious modern theatrical dance group she founded, Pappa Tarahumara. They performed around the world, and 16 years ago, had a one-night show in San Francisco. The man who would become her husband was in the audience. She soon joined him in San Francisco, often returning to Tokyo to rehearse. But after the earthquake and tsunami hit Japan in 2011, her company disbanded. . . . Grady focused on nourishing her family and creating a line of fermented products that she sells online, at local Bay Area stores and at the Ferry Plaza Farmers Market. She began at La Cocina in 2012 and named her business after her son, Aedan. (Written in Kanji characters, the name means “wisdom” and “handed down from generation to generation.”) . . . Her fermented products are “also full of wisdom about how to relate to nature and how to create a healthy life,” she says. . . .

Shani Jones, Peaches Patties

- 10 Shani Jones is a native San Franciscan. Her father was born in New Orleans, her mother, in Jamaica. She says their home was always filled with a variety of spices and dishes like jerk chicken and Jamaican patties—savory pastries filled with beef or chicken. Jones says she learned to cook from her mother, whose nickname was Peaches. After returning from college in Atlanta, Jones worked on her doctorate in organizational leadership and management while driving a Lyft⁵ car. She often told passengers about her idea of opening a catering company with her mom’s recipes. They steered her to La Cocina, where she developed her own business, named after her mother. Five years later, she caters and runs a kiosk⁶ at a small food cooperative in Bernal Heights, where some of her handmade patties have an Ethiopian twist, “because my husband is Ethiopian.” Jones has big aspirations for Peaches Patties: “The ultimate goal,” she says, “is to be the patty kingpin⁷ of the West Coast.”

From “The Food Business Incubator That Helps Immigrant Women Pursue The American Dream” by Mandalit Del Barco from WEEKEND EDITION SUNDAY, June 19, 2019. Copyright © 2019 NPR.

⁵**Lyft:** ride-sharing company where people use their own cars to provide rides to customers

⁶**kiosk:** cart or small structure with one or more open sides that is used to vend merchandise or services

⁷**kingpin:** the leader in a group or undertaking

QUALIFICATIONS FOR LA COCINA APPLICANTS

Level of Income	La Cocina considers assets and access to opportunity when screening applicants. La Cocina’s mission is to support entrepreneurs who face barriers to becoming successful in the food industry. Applicants must be classified as low- or very-low-income earners according to the Housing and Urban Development (HUD) Guidelines for San Francisco.
Business Plan	Applicants must have a business plan that is ready to be put into action. La Cocina will help applicants refine an existing business plan. Applicants without a business plan should contact one of La Cocina’s partner organizations for guidance in developing one.
Viable Product	The applicant’s food must not only be delicious but also stand out. For a business to make a profit, there also has to be a market for the food.
Entrepreneurial Work Ethic	La Cocina is excited to work with entrepreneurs who are motivated and passionate about growing their business. Succeeding in the food industry requires a lot of heart and hard work.
Community Spirit	The 30+ entrepreneurs who share La Cocina’s kitchen are from many different backgrounds. La Cocina values the community it has created and expects each entrepreneur to be active in bringing the spirit of La Cocina to the wider world. In order to succeed in the La Cocina kitchen, accepted applicants must be flexible and willing to work closely with others.

Source: La Cocina

10. Read this sentence from paragraph 2.

Two alumnae of its culinary program . . . were even recognized as semi-finalists for prestigious James Beard awards.

The words “recognized” and “prestigious” in the sentence convey that La Cocina

- E.** makes an effort to identify chefs who have a strong entrepreneurial work ethic.
- F.** has the ability to mentor chefs to reach the highest levels of achievement in their craft.
- G.** is focused on establishing local food entrepreneurs as celebrated professionals in their industry.
- H.** is dedicated to assisting chefs in developing new and innovative products.

11. Paragraph 4 contributes to the development of ideas about La Cocina in paragraph 1 by
- A. specifying how La Cocina helps local food entrepreneurs become successful.
 - B. explaining what kind of goals La Cocina tries to set for local food entrepreneurs.
 - C. providing background information about local food entrepreneurs in San Francisco.
 - D. describing how Caleb Zigas discovered the need to provide assistance to local food entrepreneurs.
12. The quotation from Caleb Zigas in paragraph 6 conveys a central idea of the excerpt by
- E. showing appreciation for the vibrant diversity of La Cocina.
 - F. communicating La Cocina’s goal of helping people of varying abilities.
 - G. describing a typical scenario at La Cocina.
 - H. emphasizing La Cocina’s focus on creating products.
13. The description of Mariko Grady’s background in paragraph 9 contributes to a central idea in the excerpt by
- A. demonstrating what people are looking for when they come to La Cocina.
 - B. explaining why La Cocina is the best option for people starting out in the food business.
 - C. showing how La Cocina can help people find success in the food business even though they have a background in something different.
 - D. confirming that La Cocina was originally established for small business owners who are looking to expand their services.
14. Which statement would the author **most likely** agree with?
- E. The publicity efforts La Cocina engages in will help its business design spread to other cities.
 - F. The variety of exciting food available in San Francisco is due to the work of La Cocina.
 - G. La Cocina offers a method that will guarantee business ownership for qualified entrepreneurs.
 - H. La Cocina does admirable work in supporting the efforts of ambitious cooks who may have few resources.

15. Which piece of evidence from the excerpt is **most** relevant to the author’s claim that La Cocina provides an opportunity to become more established as a business owner?
- A. the description of how two alumni of the business incubator have won food awards
 - B. the reference to a storytelling project where chefs relate their own backgrounds and business experiences
 - C. the lists of necessities such as business plans and city permits that La Cocina assists with
 - D. the notations of the criteria that applicants must meet to be endorsed by La Cocina in order to develop a small business
16. Which detail would be **most** important to include in a summary of the excerpt?
- E. La Cocina produced a cookbook that includes some members’ stories.
 - F. La Cocina provides legal and technical assistance to its members.
 - G. Potential vendors need a market for their food before considering La Cocina.
 - H. There are different types of venues that food vendors from La Cocina use.
17. The table provides additional support for a central idea of the excerpt by
- A. showing how La Cocina works with partner organizations to further educate and advance successful applicants within the food industry.
 - B. explaining how applicants are prepared to develop their own businesses once they leave La Cocina.
 - C. establishing that La Cocina is dedicated to its mission to provide delicious food to different communities.
 - D. emphasizing that applicants must be willing to dedicate time to promoting and developing their business with La Cocina.

CONTINUE TO THE NEXT PAGE ►

Excerpt from *In Search of the Unknown*

by Robert W. Chambers

- 1 It was at that time the policy of the trustees and officers of the Zoological Gardens neither to employ collectors nor to send out expeditions in search of specimens. The society decided to depend upon voluntary contributions, and I was always busy, part of the day, in dictating answers to correspondents who wrote offering their services as hunters of big game, collectors of all sorts of fauna, trappers, snarers, and also to those who offered specimens for sale, usually at exorbitant rates.
- 2 To the proprietors of . . . mangy lynxes, moth-eaten coyotes, and dancing bears I returned courteous but uncompromising refusals—of course, first submitting all such letters, together with my replies, to Professor Farrago.
- 3 One day towards the end of May, however, just as I was leaving Bronx Park to return to town, Professor Lesard, of the reptilian department, called out to me that Professor Farrago wanted to see me a moment; so I . . . retraced my steps to the temporary, wooden building occupied by Professor Farrago, general superintendent of the Zoological Gardens. The professor, who was sitting at his desk before a pile of letters and replies submitted for approval by me, pushed his glasses down and looked over them at me with a whimsical smile that suggested amusement, impatience, annoyance, and perhaps a faint trace of apology.
- 4 “Now, here’s a letter,” he said, with a deliberate gesture towards a sheet of paper impaled on a file—“a letter that I suppose you remember.” He disengaged the sheet of paper and handed it to me.
- 5 “Oh yes,” I replied, with a shrug; “of course the man is mistaken—or—”
- 6 “Or what?” demanded Professor Farrago, tranquilly, wiping his glasses.
- 7 “—Or a liar,” I replied.
- 8 After a silence he leaned back in his chair and bade me read the letter to him again, and I did so with a contemptuous tolerance for the writer, who must have been either a very innocent victim or a very stupid swindler. I said as much to Professor Farrago, but, to my surprise, he appeared to waver.
- 9 “I suppose,” he said, with his near-sighted, embarrassed smile, “that nine hundred and ninety-nine men in a thousand would throw that letter aside and condemn the writer as a liar or a fool?”
- 10 “In my opinion,” said I, “he’s one or the other.”
- 11 “He isn’t—in mine,” said the professor, placidly.
- 12 “What!” I exclaimed. “Here is a man living all alone on a strip of rock and sand between the wilderness and the sea, who wants you to send somebody to take charge of a bird that doesn’t exist!”

- 13 “How do you know,” asked Professor Farrago, “that the bird in question does not exist?”
- 14 “It is generally accepted,” I replied, sarcastically, “that the great auk has been extinct for years. Therefore I may be pardoned for doubting that our correspondent possesses a pair of them alive.”
- 15 “Oh, you young fellows,” said the professor, smiling wearily, “you embark on a theory for destinations that don’t exist.”
- 16 He leaned back in his chair, his amused eyes searching space for the imagery that made him smile.
- 17 “Like swimming squirrels, you navigate with the help of Heaven and a stiff breeze, but you never land where you hope to—do you?”
- 18 Rather red in the face, I said: “Don’t you believe the great auk to be extinct?”
- 19 “Audubon¹ saw the great auk.”
- 20 “Who has seen a single specimen since?”
- 21 “Nobody—except our correspondent here,” he replied, laughing.
- 22 I laughed, too, considering the interview at an end, but the professor went on, coolly:
- 23 “Whatever it is that our correspondent has—and I am daring to believe that it *is* the great auk itself—I want you to secure it for the society.”
- 24 When my astonishment subsided my first conscious sentiment was one of pity. Clearly, Professor Farrago was on the verge of dotage²—ah, what a loss to the world!
- 25 I believe now that Professor Farrago perfectly interpreted my thoughts, but he betrayed neither resentment nor impatience. I drew a chair up beside his desk—there was nothing to do but to obey, and this fool’s errand was none of my conceiving.
- 26 Together we made out a list of articles necessary for me and itemized the expenses I might incur, and I set a date for my return, allowing no margin for a successful termination to the expedition.
- 27 “Never mind that,” said the professor. “What I want you to do is to get those birds here safely. Now, how many men will you take?”
- 28 “None,” I replied, bluntly; “it’s a useless expense, unless there is something to bring back. If there is I’ll wire you, you may be sure.”
- 29 “Very well,” said Professor Farrago, good-humoredly, “you shall have all the assistance you may require. Can you leave to-night?”

¹**Audubon:** John James Audubon, an ornithologist and artist who created scientific illustrations of birds

²**dotage:** a loss of reasoning brought about by old age

- 30 The old gentleman was certainly prompt. I nodded, half-sulkily, aware of his amusement.
- 31 “So,” I said, picking up my hat, “I am to start north to find a place called Black Harbor, where there is a man named Halyard who possesses, among other household utensils, two extinct great auks—”
- 32 We were both laughing by this time. I asked him why on earth he credited the assertion of a man he had never before heard of.
- 33 “I suppose,” he replied, with the same half-apologetic, half-humorous smile, “it is instinct. I feel, somehow, that this man Halyard *has* got an auk—perhaps two. I can’t get away from the idea that we are on the eve of acquiring the rarest of living creatures. It’s odd for a scientist to talk as I do; doubtless you’re shocked—admit it, now!”
- 34 But I was not shocked; on the contrary, I was conscious that the same strange hope that Professor Farrago cherished was beginning, in spite of me, to stir my pulses, too.
- 35 “If he has—” I began, then stopped.
- 36 The professor and I looked hard at each other in silence.
- 37 “Go on,” he said, encouragingly.
- 38 But I had nothing more to say, for the prospect of beholding with my own eyes a living specimen of the great auk produced a series of conflicting emotions within me which rendered speech profanely superfluous.

From IN SEARCH OF THE UNKNOWN by Robert W. Chambers—Public Domain

18. Read paragraph 2 from the excerpt.

To the proprietors of . . . mangy lynxes, moth-eaten coyotes, and dancing bears I returned courteous but uncompromising refusals—of course, first submitting all such letters, together with my replies, to Professor Farrago.

This paragraph helps develop the plot by establishing that the narrator

- E. dislikes writing refusal letters for the animals offered to the zoological society.
- F. attempts to predict what the professor would say in the refusal letters.
- G. believes that many of the animals offered are not acceptable for the zoological society.
- H. resents the professor’s insistence on reviewing the refusal letters.

19. Read this sentence from paragraph 3.

The professor, who was sitting at his desk before a pile of letters and replies submitted for approval by me, pushed his glasses down and looked over them at me with a whimsical smile that suggested amusement, impatience, annoyance, and perhaps a faint trace of apology.

What does the phrase “a faint trace of apology” convey about the professor?

- A. It indicates that the professor feels bad that he has to call the narrator to his office after work.
 - B. It shows that the professor is hesitant to share his opinions with the narrator.
 - C. It implies that the professor is uncomfortable criticizing the narrator’s work.
 - D. It suggests that the professor knows that the conversation will be frustrating for the narrator.
20. How does the exchange between the professor and the narrator in paragraphs 8–11 contribute to the development of the characters?
- E. It establishes the conflict between the professor and the narrator concerning the validity of the letter.
 - F. It suggests a theme of collaboration because the narrator and the professor regularly work together.
 - G. It reveals the characters’ traits by contrasting the narrator’s distrust with how easily the professor is deceived by what he reads.
 - H. It hints that the resolution will involve the narrator accepting the professor’s opinion about the content of the letter.
21. The professor’s observations in paragraphs 15–17 create tension in the excerpt by causing the narrator to feel
- A. flustered by the professor’s criticism of his logic.
 - B. annoyed by the professor’s sarcasm about his inexperience.
 - C. confused by the professor’s lack of respect for his opinion.
 - D. frustrated by the professor’s lack of interest in his theory.

22. How does the interaction between the narrator and the professor in paragraphs 26–28 contribute to the development of the theme?
- E. It illustrates the professor’s patience as the narrator argues against making the expedition.
 - F. It reveals the narrator’s frustration with his limited role in making decisions for the zoological society.
 - G. It emphasizes the professor’s desire to acquire new specimens for the zoological society at any cost.
 - H. It shows the narrator’s acceptance of his assignment despite his personal objections.
23. Which sentence from the excerpt **best** explains why the professor is eager to send the narrator on an expedition?
- A. “I believe now that Professor Farrago perfectly interpreted my thoughts, but he betrayed neither resentment nor impatience.” (paragraph 25)
 - B. “Together we made out a list of articles necessary for me and itemized the expenses I might incur, and I set a date for my return, allowing no margin for a successful termination to the expedition.” (paragraph 26)
 - C. “ ‘What I want you to do is to get those birds here safely.’ ” (paragraph 27)
 - D. “ ‘I can’t get away from the idea that we are on the eve of acquiring the rarest of living creatures.’ ” (paragraph 33)
24. How does paragraph 34 help develop the plot of the excerpt?
- E. It shows that the narrator is beginning to consider the possibility of finding the great auks.
 - F. It demonstrates that the narrator is struggling to understand why the professor thinks the great auks exist.
 - G. It establishes that the narrator is willing to let the professor overrule him about the great auks.
 - H. It emphasizes that the narrator feels a sense of urgency to complete the expedition to locate the great auks.

- 25.** Which sentence **best** demonstrates the professional relationship between the narrator and the professor?
- A.** "He disengaged the sheet of paper and handed it to me." (paragraph 4)
 - B.** "Clearly, Professor Farrago was on the verge of dotage—ah, what a loss to the world!" (paragraph 24)
 - C.** "I drew a chair up beside his desk—there was nothing to do but to obey, and this fool's errand was none of my conceiving." (paragraph 25)
 - D.** " 'Very well,' said Professor Farrago, good-humoredly, 'you shall have all the assistance you may require.' " (paragraph 29)
- 26.** How does the author develop the contrast between the narrator's point of view and the professor's point of view?
- E.** by providing both the narrator's and professor's thoughts on how age and experience influence each other's reasoning
 - F.** by using the conversation between the narrator and the professor to emphasize their reactions to the letter
 - G.** by describing the professor's persistent efforts to change the narrator's mind about the letter
 - H.** by including dialogue that explains why the professor is the supervisor and the narrator is his subordinate

In 1903 brothers Wilbur and Orville Wright conducted experiments related to flying machines. These experiments would eventually lead to air travel becoming a reliable form of transportation.

Excerpt from “How We Made the First Flight”

by Orville Wright

- 1 During the night of December 16, 1903, a strong cold wind blew from the north. When we arose on the morning of the 17th, the puddles of water, which had been standing about camp since the recent rains, were covered with ice. The wind had a velocity of 10 to 12 meters per second (22 to 27 miles an hour). We thought it would die down before long, and so remained indoors the early part of the morning. But when ten o'clock arrived, and the wind was as brisk as ever, we decided that we had better get the machine out and attempt a flight. We hung out the signal for the men of the Life Saving Station.¹ We thought that by facing the flyer into a strong wind, there ought to be no trouble in launching it from the level ground about camp. We realized the difficulties of flying in so high a wind, but estimated that the added dangers in flight would be partly compensated for by the slower speed in landing.

Final Preparations

- 2 We laid the track on a smooth stretch of ground about one hundred feet north of the new building. The biting cold wind made work difficult, and we had to warm up frequently in our living room, where we had a good fire in an improvised stove made of a large carbide² can. By the time all was ready, J. T. Daniels, W. S. Dough and A. D. Etheridge, members of the Kill Devil³ Life Saving Station; W. C. Brinkley of Manteo, and Johnny Moore, a boy from Nags Head,⁴ had arrived.
- 3 We had a “Richard” hand anemometer⁵ with which we measured the velocity of the wind. Measurements made just before starting the first flight showed velocities of 11 to 12 meters per second, or 24 to 27 miles per hour. . . .

Audacity—and Calculation

- 4 Wilbur having used his turn in the unsuccessful attempt on the 14th, the right to the first trial now belonged to me. After running the motor a few minutes to heat it up, I released the wire that held the machine to the track, and the machine started forward in the wind. Wilbur ran at the side of the machine, holding the wing to balance it on the track. Unlike the start on the 14th, made in a calm, the machine, facing a 27-mile wind, started very slowly. Wilbur was able to stay with it till it lifted from the track after a forty-foot run. One of the Life Saving men snapped the camera for us, taking a picture just as the machine had reached the end of the track and had

¹**Life Saving Station:** one of the rescue stations along the Atlantic coastline that provided assistance to mariners in distress

²**carbide:** a very hard material composed of carbon and other heavy metals

³**Kill Devil:** the town of Kill Devil Hills in eastern North Carolina

⁴**Nags Head:** a town in eastern North Carolina

⁵**“Richard” hand anemometer:** the type of anemometer produced by Jules Richard, a French manufacturer of scientific instruments

risen to a height of about two feet. The slow forward speed of the machine over the ground is clearly shown in the picture by Wilbur's attitude. He stayed along beside the machine without any effort.

Flight

- 5 The course of the flight up and down was exceedingly erratic, partly due to the irregularity of the air, and partly to lack of experience in handling this machine. The control of the front rudder was difficult on account of its being balanced too near the center. This gave it a tendency to turn itself when started; so that it turned too far on one side and then too far on the other. As a result the machine would rise suddenly to about ten feet, and then as suddenly dart for the ground. A sudden dart when a little over a hundred feet from the end of the track, or a little over 120 feet from the point at which it rose into the air, ended the flight. As the velocity of the wind was over 35 feet per second and the speed of the machine over the ground against this wind ten feet per second, the speed of the machine relative to the air was over 45 feet per second, and the length of the flight was equivalent to a flight of 540 feet made in calm air. This flight lasted only 12 seconds, but it was nevertheless the first in the history of the world in which a machine carrying a man had raised itself by its own power into the air in full flight, had sailed forward without reduction of speed and had finally landed at a point as high as that from which it started.

From "How We Made the First Flight" by Orville Wright—Public Domain/Federal Aviation Administration

- 27.** How does paragraph 1 introduce the idea that the Wright brothers knew that their flight attempt was risky?
- A.** through the mention of a signal to notify lifesaving experts that the flight attempt was about to begin
 - B.** by providing specific details about the speed of the wind and the Wright brothers' response to the windy conditions
 - C.** by suggesting that a slower landing would be necessary at the end of the flight in order to maintain safety
 - D.** through the indication that the Wright brothers waited indoors for most of the morning because of the poor weather

28. Read this sentence from paragraph 4.

Wilbur having used his turn in the unsuccessful attempt on the 14th, the right to the first trial now belonged to me.

The sentence contributes to the development of ideas in the excerpt by

- E.** demonstrating the challenge of the extreme winter conditions during the flight.
- F.** revealing how many tries it took for Wilbur Wright to finally get the machine to take flight.
- G.** demonstrating that both Orville and Wilbur Wright were eager to pilot what could potentially be the first flight.
- H.** suggesting a sense that both brothers felt confident they would soon succeed in completing the first flight.

29. The photograph mentioned in paragraph 4 is significant because it

- A.** documents that the machine rose into the air as soon as the tethering wire was released.
- B.** provides proof of the critical moment the machine took flight.
- C.** documents that the wind reduced the speed of the plane at the start of the flight.
- D.** provides proof that the pilot had to gradually increase the height of the plane in the air.

30. How do the details in paragraph 5 about the uneven nature of the flight convey a central idea of the excerpt?

- E.** by explaining how the flawed design of the machine caused it to turn unpredictably in the air and brought the first flight by a person to an abrupt end
- F.** by indicating that the difficulty in controlling the flight was caused by the rudimentary instruments of the machine and the inexperience of the pilot
- G.** by explaining how the pilot and the plane overcame adverse conditions in order to complete the first piloted flight
- H.** by indicating that the gradual change in wind velocity created an extreme environment in which to maneuver the plane and maintain its flight

31. Read this sentence from paragraph 5.

As the velocity of the wind was over 35 feet per second and the speed of the machine over the ground against this wind ten feet per second, the speed of the machine relative to the air was over 45 feet per second, and the length of the flight was equivalent to a flight of 540 feet made in calm air.

How does the sentence help convey Orville Wright's perspective about this first flight?

- A. It suggests that he was frustrated by the poor flying conditions on the day of the flight.
- B. It emphasizes that he believed the flight was successful despite its short distance.
- C. It provides a comparison between flight distances under calm and high wind conditions.
- D. It highlights the importance of such calculations in the success of future flights.

32. Read this sentence from paragraph 5.

A sudden dart when a little over a hundred feet from the end of the track, or a little over 120 feet from the point at which it rose into the air, ended the flight.

How does the sentence contribute to the paragraph?

- E. It details the need for the pilot to have quick reflexes while flying the machine.
- F. It presents the idea that the difficulty of operating the machine shortened the flight.
- G. It describes the shift in wind speed that made flying nearly impossible.
- H. It provides an overview of the flight's progression from takeoff to landing.

33. Read this sentence from paragraph 5.

This flight lasted only 12 seconds, but it was nevertheless the first in the history of the world in which a machine carrying a man had raised itself by its own power into the air in full flight, had sailed forward without reduction of speed and had finally landed at a point as high as that from which it started.

The words “only,” “nevertheless,” and “finally” **most** clearly convey the idea that

- A. even a flight of such minor duration had taken a long time to achieve.
 - B. the short flight gave the Wright brothers hope for longer ones in the future.
 - C. the flight proved that the machine was, at last, capable of becoming airborne.
 - D. although it was brief, the flight was a remarkable accomplishment.
34. Which sentence from the excerpt **best** supports the idea that the Wright brothers had to adapt their plans for the flight in order to accommodate the weather conditions?
- E. “When we arose on the morning of the 17th, the puddles of water, which had been standing about camp since the recent rains, were covered with ice.” (paragraph 1)
 - F. “We realized the difficulties of flying in so high a wind, but estimated that the added dangers in flight would be partly compensated for by the slower speed in landing.” (paragraph 1)
 - G. “After running the motor a few minutes to heat it up, I released the wire that held the machine to the track, and the machine started forward in the wind.” (paragraph 4)
 - H. “The course of the flight up and down was exceedingly erratic, partly due to the irregularity of the air, and partly to lack of experience in handling this machine.” (paragraph 5)
35. The use of chronological structure contributes to the development of ideas in the excerpt by
- A. outlining the actions that the Wright brothers took to prepare for and successfully complete the first flight.
 - B. identifying the primary factors that allowed the Wright brothers to overcome obstacles and achieve the first flight.
 - C. showing how the Wright brothers applied lessons learned from their previous flight attempts to accomplish the first flight.
 - D. demonstrating how the Wright brothers analyzed the impact of wind velocity to identify the ideal conditions for the first flight.

CONTINUE TO THE NEXT PAGE ►

Cross-Purposes

What I am is *built*: concrete and steel.
I defy gravity. I am what every athlete
wants: to remain at the apex of the leap,
up in the air. And yet I am useful, too:

5 cars, trucks, people, even trains
make their way across my broad back.
Swallows and ospreys¹ nest in my trusses.

10 *What I am is motion. I am water, and I am older
than anything else you know. No human
built me. I am gravity's best friend; I pool
and flow wherever gravity takes me.
I am the blood flowing in the runner's chest,
and I catch everything: from the hills,
the mountains. It all washes down through me.*

15 What you are is an *accident*,
what happens to rain when rain gives in
to Earth's gravitational pull.
You are some tears dribbling from a mountain's
eye, running down the pavements
20 of small towns, into the cities, to the sea.
You are the path of least resistance.

25 *What I am is power. You, of course,
have none: you are a static lump, an artifact
slowly decaying. But my regal flow
nourishes grasses, permits empires to rise.
Those who made you will break you,
in time, replacing you with yet another
clumsy structure. I have seen. I know.*

"Clumsy"? Being rebuilt makes me
30 a friend of time, does it not? And it means
that I have siblings—those "clumsy" structures,
my sisters and brothers.
We stitch across the rip you make.
We are steel thread to the human needle.
35 We bind you up. We sew you.

¹ospreys: large birds

40

*And I sow into you; in every cranny
of your superstructure my vapors cling.
They bring out your softness, your rust.
Boast your best, and boast better yet.
I am listening to the bright hum
of the wind in your wires. Because I am,
above all else, patient. I will wait for you.*

- 36.** How does the similar construction of line 1 and line 8 contribute to the meaning of the poem?
- E.** It introduces the intended permanence of the structure and the ever-changing fluidity of the water.
 - F.** It shows that the structure can bridge the gap caused by the water.
 - G.** It suggests that the inflexible structure has more limitations than the adaptable water does.
 - H.** It contrasts the stability of the structure with the instability of the water.
- 37.** Read lines 2–4 and lines 12–14 from the poem.

**I am what every athlete
wants: to remain at the apex of the leap,
up in the air.**

***I am the blood flowing in the runner's chest,
and I catch everything: from the hills,
the mountains.***

How do the lines contribute to the development of a central idea of the poem?

- A.** They establish that both the structure and the water have endurance and control.
- B.** They highlight that both the structure and the water are powerful and impressive.
- C.** They suggest that the structure and the water are unaware of how similar they are.
- D.** They reveal that the structure and the water are surprised that they are interrelated.

38. The use of the words “siblings” and “my sisters and brothers” in lines 31–32 conveys the idea that the
- E. forms water can take are less diverse than the types of structures that exist.
 - F. number of human-made structures is rapidly increasing.
 - G. water passes under many similar-looking structures as it flows.
 - H. structure is powerful because it is one of many.
39. The comparison to sewing in lines 33–35 helps show that the structure
- A. enhances the beauty of the natural landscape.
 - B. brings people together more effectively than nature does.
 - C. provides clear boundaries for natural environments.
 - D. serves as a means for people to overcome an obstacle created by nature.
40. The last stanza (lines 36–42) conveys a central idea of the poem by
- E. demonstrating that both the structure and the water depend on each other to fulfill their functions.
 - F. implying that a stronger structure would be able to resist the degradation caused by the water.
 - G. revealing that the passage of time will render both the structure and the water obsolete.
 - H. suggesting that the water will eventually weaken the structure and will continue to exist after the structure is gone.
41. Read lines 41–42 from the poem.

***Because I am,
above all else, patient. I will wait for you.***

Which of the following supports what is implied in these lines?

- A. “I am older / than anything else you know.” (lines 8–9)
- B. “No human / built me.” (lines 9–10)
- C. “It all washes down through me.” (line 14)
- D. “Those who made you will break you,” (line 26)

- 42.** How does the poet develop the points of view of the structure and the water?
- E.** by giving an account of a discussion between them about the future of human civilization
 - F.** by narrating a debate they have over their impact on the environment
 - G.** by illustrating the unique power they each possess over nature
 - H.** by using personification to allow them to debate who is more important
- 43.** How does the form of the poem contribute to its meaning?
- A.** The use of an equal number of lines in each stanza emphasizes that both speakers are equally important.
 - B.** The use of italics in some of the stanzas indicates the increasing tension between the structure and the water.
 - C.** The alternating positions of the stanzas highlights the opposing points of view of the speakers.
 - D.** The lack of a regular rhyme scheme or meter reflects the way the water changes the structure and the way the water itself changes.

Excerpt from “Scribe Like an Egyptian”

by Hilary Wilson

- 1 In ancient Egypt, literacy was the key to success. However, contrary to popular belief, not all Egyptian scribes understood hieroglyphs. Many relied instead on the simpler hieratic script for the multitude of everyday documents generated by the Egyptian bureaucracy.¹
- 2 Hieroglyphs—“the Words of God”—compose a writing system with more than 1,000 distinct characters, the meanings of which were lost for 1,500 years before they were deciphered by Jean-François Champollion in 1822. Including both ideograms (which convey a whole word or idea, either concrete or abstract, in a single sign) and phonograms (representing either an alphabetic sound or a group of consonants), [the writing system] was used in formal inscriptions on tomb and temple walls as well as on elaborate funerary papyri.² For everyday purposes, however, scribes used a shorthand version of the hieroglyphic script known as hieratic, which was quicker to write and more economical of space. The two writings existed side by side for at least 2,500 years.
- 3 Scraps of ancient hieratic writing, mostly penned by student scribes on limestone flakes called ostraca, suggest that no matter how humble his origins, an educated Egyptian could achieve almost anything. Horemheb (d.1292 B.C.) is a good example. Born of middle-ranking parents, his scribal training led to an army career. From Scribe of Recruits, during the reign of Akhenaten (1353–1336 B.C.), Horemheb rose through the military ranks and, by the rule of Tutankhamun (1332–1323 B.C.), he was commander in chief of the Egyptian forces. As a close adviser of the young Pharaoh, Horemheb was appointed “Deputy of the King throughout the Two Lands,” and might have expected to succeed to the throne should the king die childless. He had to wait a few years, but eventually Horemheb achieved the pinnacle of his career by becoming the last king of the 18th Dynasty, making his mark by instituting dramatic reforms to the organisation of the army, the judiciary and administration in general. The lasting success of these changes owed much to his scribal background. . . .
- 4 But education was not available to all. Government departments and major temples supported schools, where boys commenced their training at six or seven, sometimes earlier. To these boarding establishments . . . family or household servants delivered the students’ food and drink rations daily for several years, during which time the student was not contributing to the family’s income. Boys from poorer families could only hope to be educated with support from a wealthier relative or patron, or through apprenticeship to an older scribe, perhaps the local clerk or land agent, who would teach them the basics of the scribe’s craft. This limited the scope for employment but such “on the job” training allowed apprentices to help out at home while learning. . . .
- 5 Scribal education began with the elementary principles of the hieratic script. The lowliest scribes, who trained for just five or six years, probably learned only the rudiments³ of the hieroglyphic

¹**bureaucracy:** an administrative staff of government officials

²**funerary papyri:** a sheet or scroll of papyrus containing religious images and hieroglyphs meant to help the deceased be reborn in the afterlife

³**rudiments:** basics

script. Students were set exemplar documents and extracts from popular texts to copy, to practice their hieratic handwriting on basic-format letters, reports and contracts, while absorbing the good advice contained in the texts. Surviving examples of copy-work sometimes include tutors' corrections added in red. Some significant Egyptian literary works survive almost exclusively from student copies.

- 6 A schoolboy⁴ "dictionary" of hieroglyphs with their hieratic equivalents shows that a knowledge of more than 450 signs was required for everyday writing purposes. Lessons in record-keeping and filing and labelling enabled any half-competent scribe to perform that most essential of all scribal functions: the making and updating of lists. For professions such as those of government official, priest or lawyer, a scribe would train for several more years, increasing his vocabulary to perhaps a thousand or more signs. Those with the best handwriting or drawing skills might follow the craft of creating beautifully illustrated copies of funerary texts, commonly called *Books of the Dead*. Others could become draughtsmen,⁵ artists or architects. Doctors compiled their own collections of medication recipes, treatments and associated incantations, many copied from texts found in the House of Life, the temple library. Lawyers had to be familiar with the corpus⁶ of civil and religious laws and precedents found in the official records, which were administered by archivists. Egypt's bureaucratic society depended on the skills of an army of scribes of all ranks from filing clerk to tax assessor. For young Egyptians, "be a scribe" was the best of career advice.

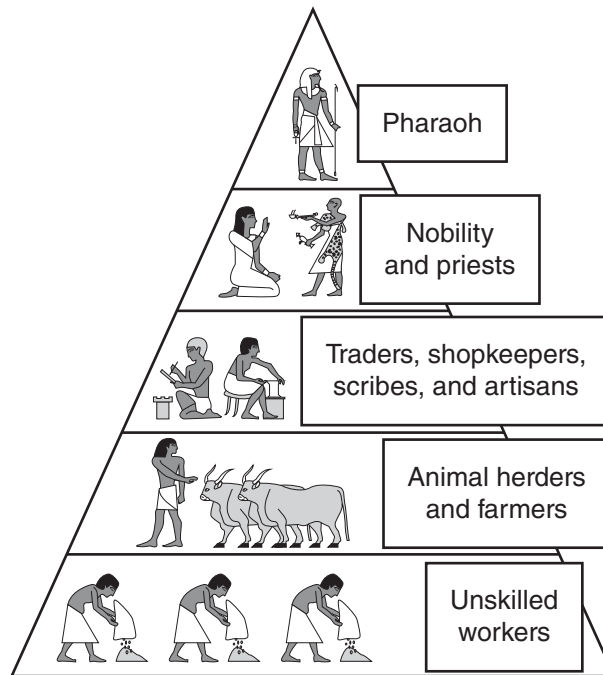
From "Scribe Like an Egyptian" by Hilary Wilson from HISTORY TODAY, August 8, 2019. Copyright © 2019 by History Today Ltd. Company.

⁴**schoolboy:** slang that refers to materials used during the course of receiving an education

⁵**draughtsmen:** people who draw plans of machinery or structures

⁶**corpus:** a collection of writings

ANCIENT EGYPTIAN
SOCIAL CLASS STRUCTURE



44. Which sentence from the excerpt **best** supports the idea that there were different levels of education for a scribe?
- E. "However, contrary to popular belief, not all Egyptian scribes understood hieroglyphs." (paragraph 1)
 - F. "For everyday purposes, however, scribes used a shorthand version of the hieroglyphic script known as hieratic, which was quicker to write and more economical of space." (paragraph 2)
 - G. "This limited the scope for employment but such 'on the job' training allowed apprentices to help out at home while learning." (paragraph 4)
 - H. "For professions such as those of government official, priest or lawyer, a scribe would train for several more years, increasing his vocabulary to perhaps a thousand or more signs." (paragraph 6)

45. Read this sentence from paragraph 2.

Including both ideograms (which convey a whole word or idea, either concrete or abstract, in a single sign) and phonograms (representing either an alphabetic sound or a group of consonants), [the writing system] was used in formal inscriptions on tomb and temple walls as well as on elaborate funerary papyri.

The author's purpose for including the sentence is to

- A. explain that hieroglyphs were reserved for royal and religious purposes.
- B. demonstrate the methods experts use to interpret hieroglyphs.
- C. clarify the significance of hieroglyphs in language and literature.
- D. describe the basic features of hieroglyphs and how they were typically used.

46. Which statement summarizes the process that schools used to train scribes?

- E. Boys had to memorize around 450 hieratic signs. Once they had learned these, they were expected to copy literary texts that contained valuable lessons.
- F. Boys spent at least five years learning hieratic signs, which they practiced by copying texts. During this period, they were also introduced to basic hieroglyphs.
- G. Boys spent five or six years learning the basics of hieratic script. After this period, they could become apprentices and begin learning hieroglyphics.
- H. Boys were taught two different scripts. They practiced both types by copying lists and eventually progressed to writing letters, reports, and contracts.

47. Which claim is **best** supported by the information presented in paragraphs 3 and 4?

- A. Education was difficult for the common citizens of Egypt to obtain.
- B. Once students began studying to become scribes, their future held more possibilities.
- C. If someone wanted to become a scribe, support from outside sources such as the government or a temple was necessary.
- D. Becoming a successful scribe required a student's family to sacrifice time and money in order to ensure the best education.

48. Hieratic script was particularly valuable in a bureaucratic government because
- E. using hieratic script was an efficient way for scribes to produce large amounts of documentation.
 - F. many members of the general public could comprehend information that was written in hieratic script.
 - G. knowledge of hieratic script prepared scribes to serve their society in increasingly challenging roles.
 - H. people from different professional backgrounds could easily communicate with one another in hieratic script.
49. The diagram after paragraph 6 **best** provides additional support for the topic of the excerpt by
- A. demonstrating the place of scribes in ancient Egyptian social structure.
 - B. depicting the types of tasks performed by each social class.
 - C. indicating that scribes were able to easily improve their social ranking.
 - D. comparing the social positions of each social class in ancient Egypt.
50. Based on the excerpt, which statement would the author **most** strongly agree with?
- E. Boys in Egypt were encouraged to enter the same profession as older male relatives.
 - F. Knowledge of hieroglyphs was helpful to Egyptians who were interested in becoming scribes.
 - G. Hieratic script was more practical to ancient Egyptians than hieroglyphs were.
 - H. Boys in Egypt who studied for only a few years were unlikely to find employment as scribes.

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Excerpt from “Niagara Falls”

by Rupert Brooke

- 1 The real secret of the beauty and terror of the Falls is not their height or width, but the feeling of colossal power and of unintelligible disaster caused by the plunge of that vast body of water. If that were taken away, there would be little visible change, but the heart would be gone.
- 2 The American Falls do not inspire this feeling in the same way as the Canadian. It is because they are less in volume, and because the water does not fall so much into one place. By comparison their beauty is almost delicate and fragile. They are extraordinarily level, one long curtain of lacework and woven foam. Seen from opposite, when the sun is on them, they are blindingly white, and the clouds of spray show dark against them. With both Falls the colour of the water is the ever-altering wonder. Greens and blues, purples and whites, melt into one another, fade, and come again, and change with the changing sun. Sometimes they are as richly diaphanous¹ as a precious stone, and glow from within with a deep, inexplicable light. Sometimes the white intricacies of dropping foam become opaque and creamy. And always there are the rainbows. If you come suddenly upon the Falls from above, a great double rainbow, very vivid, spanning the extent of spray from top to bottom, is the first thing you see. If you wander along the cliff opposite, a bow springs into being in the American Falls, accompanies you courteously on your walk, dwindles and dies as the mist ends, and awakens again as you reach the Canadian tumult. And the bold traveller who attempts the trip under the American Falls sees, when he dare open his eyes to anything, tiny baby rainbows, some four or five yards in span, leaping from rock to rock among the foam, and gambolling beside him, barely out of hand's reach, as he goes. One I saw in that place was a complete circle, such as I have never seen before, and so near that I could put my foot on it. It is a terrifying journey, beneath and behind the Falls. The senses are battered and bewildered by the thunder of the water and the assault of wind and spray; or rather, the sound is not of falling water, but merely of falling; a noise of unspecified ruin. So, if you are close behind the endless clamour, the sight cannot recognise liquid in the masses that hurl past. You are dimly and pitifully aware that sheets of light and darkness are falling in great curves in front of you. Dull omnipresent foam washes the face. Farther away, in the roar and hissing, clouds of spray seem literally to slide down some invisible plane of air.
- 3 Beyond the foot of the Falls the river is like a slipping floor of marble, green with veins of dirty white, made by the scum that was foam. It slides very quietly and slowly down for a mile or two, sullenly exhausted. Then it turns to a dull sage green, and hurries more swiftly, smooth and ominous. As the walls of the ravine close in, trouble stirs, and the waters boil and eddy. These are the lower rapids, a sight more terrifying than the Falls, because less intelligible. Close in its bands of rock the river surges tumultuously forward, writhing and leaping as if inspired by a demon. It is pressed by the straits into a visibly convex form. Great planes of water slide past. Sometimes it is thrown up into a pinnacle of foam higher than a house, or leaps with incredible speed from the crest of one vast wave to another, along the shining curve between, like the spring of a wild beast. Its motion continually suggests muscular action. The power manifest in these rapids moves one with a different sense of awe and terror from that of the Falls. Here the inhuman life and strength are spontaneous, active, almost resolute. . . . A place of fear.

¹**diaphanous:** sheer, translucent

4 One is drawn back, strangely, to a contemplation of the Falls, at every hour, and especially by night, when the cloud of spray becomes an immense visible ghost, straining and wavering high above the river, white and pathetic and translucent. The Victorian lies very close below the surface in every man. There one can sit and let great cloudy thoughts of destiny and the passage of empires drift through the mind; for such dreams are at home by Niagara. I could not get out of my mind the thought of a friend, who said that the rainbows over the Falls were like the arts and beauty and goodness, with regard to the stream of life—caused by it, thrown upon its spray, but unable to stay or direct or affect it, and ceasing when it ceased. In all comparisons that rise in the heart, the river, with its multitudinous waves and its single current, likens itself to a life, whether of an individual or of a community. A man's life is of many flashing moments, and yet one stream; a nation's flows through all its citizens, and yet is more than they. In such places, one is aware, with an almost insupportable and yet comforting certitude, that both men and nations are hurried onwards to their ruin or ending as inevitably as this dark flood. Some go down to it unreluctant, and meet it, like the river, not without nobility. And as incessant, as inevitable, and as unavailing as the spray that hangs over the Falls, is the white cloud of human crying. . . . With some such thoughts does the platitudinous² heart win from the confusion and thunder of a Niagara peace that the quietest plains or most stable hills can never give.

From LETTERS FROM AMERICA by Rupert Brooke—Public Domain

²**platitudinous:** clichéd, common

51. The central idea that the Falls communicate a feeling of “unintelligible disaster” (paragraph 1) is conveyed in paragraph 2 through a description of
- A. the glow of the precious stones visible within the water.
 - B. the dynamic flow of the colors that are visible in the water.
 - C. the sudden appearance and disappearance of rainbows.
 - D. the sounds associated with a sense of falling.

52. Read this sentence from paragraph 2.

They are extraordinarily level, one long curtain of lacework and woven foam.

What is the effect of comparing the American Falls to a “long curtain of lacework and woven foam”?

- E. It demonstrates the timelessness of the American Falls.
- F. It conveys the secretive nature of the American Falls.
- G. It illustrates the elegant uniformity of the American Falls.
- H. It communicates the intense strength of the American Falls.

53. Read this sentence from paragraph 3.

These are the lower rapids, a sight more terrifying than the Falls, because less intelligible.

Which statement **best** describes how the sentence fits into the overall structure of the excerpt?

- A. It signals a change from the positive aspects of the Falls to the negative aspects.
 - B. It indicates a progression from the literal description of the water to a discussion of timeless truths.
 - C. It reinforces a shift from the qualities of the Falls to the qualities of the river.
 - D. It introduces a contrast between the obvious and the hidden features of the rapids.
54. Which sentence from the excerpt **best** supports the idea that the essence of the Falls lies in their emotional impact?
- E. "If that were taken away, there would be little visible change, but the heart would be gone." (paragraph 1)
 - F. "By comparison their beauty is almost delicate and fragile." (paragraph 2)
 - G. "One is drawn back, strangely, to a contemplation of the Falls, at every hour, and especially by night, when the cloud of spray becomes an immense visible ghost, straining and wavering high above the river, white and pathetic and translucent." (paragraph 4)
 - H. "A man's life is of many flashing moments, and yet one stream; a nation's flows through all its citizens, and yet is more than they." (paragraph 4)

55. Read this sentence from paragraph 4.

There one can sit and let great cloudy thoughts of destiny and the passage of empires drift through the mind; for such dreams are at home by Niagara.

The sentence **most** contributes to the development of ideas in the excerpt by

- A. suggesting that viewing the Falls can be a life-changing experience.
- B. showing that the cliffs of the Falls are a good place for self-examination.
- C. emphasizing that the grandeur of the Falls seems impossible to grasp.
- D. highlighting the type of reflection that is inspired by a visit to the Falls.

56. In paragraph 4, the idea that human life and history ultimately end in the same way is illustrated **mainly** through
- E. the discussion of how the rainbows visible in the Falls are like the art and beauty created by humankind.
 - F. the comparison between the movement of water in the Falls and the human experience.
 - G. the inclusion of details that show that every observer's experience with the Falls is different.
 - H. the acknowledgment that contemplating the Falls at night sparks an awareness of humankind's destiny.
57. With which statement would the author of this excerpt **most likely** agree?
- A. A sense of ease and assurance comes with accepting one's fate.
 - B. No matter where one's path goes in life, one will always have regrets.
 - C. The best way to overcome fear is to recognize it and then defy it.
 - D. Reason will die out with humanity, but art will remain immortal.

PART 2 — MATHEMATICS

57 QUESTIONS

IMPORTANT NOTES

- (1) Formulas and definitions of mathematical terms and symbols are **not** provided.
- (2) Diagrams other than graphs are **not** necessarily drawn to scale. Do not assume any relationship in a diagram unless it is specifically stated or can be determined from the information given.
- (3) Assume that a diagram is in one plane unless the question specifically states that it is not.
- (4) Graphs are drawn to scale. Unless stated otherwise, you can assume relationships according to appearance. For example, lines on a graph that appear to be parallel can be assumed to be parallel. This is also true for concurrent lines, straight lines, collinear points, right angles, etc.
- (5) Reduce (simplify) all fractions to lowest terms.

GRID-IN QUESTION NOTES

- (1) For each grid-in question, write your answer at the top of the grid.
- (2) Begin recording your answer in the columns on the far left.
- (3) Fill in the circle under the box that matches the number or symbol you wrote. Leave the negative sign bubble blank if your answer is positive.

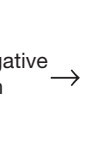
(Answer: -1.5)



Negative sign →

← Decimal point

(Answer: 3.2)



Negative sign →

← Decimal point

CONTINUE TO THE NEXT PAGE ►

GRID-IN QUESTIONS

QUESTIONS 58–62

DIRECTIONS: Solve each problem. On the answer sheet, write your answer in the boxes at the top of the grid. Start on the left side of each grid. Print only one number or symbol in each box. Under each box, fill in the circle that matches the number or symbol you wrote above.

- Do not fill in a circle under an unused box.
 - Do not leave a box blank in the middle of an answer.
-

58.

LAST YEAR'S TOTAL SALES

Vehicles	Number
Four-door cars	5,850
Two-door cars	2,250
Trucks	900
Total	9,000

The table above shows the number of each type of vehicle sold at a dealership last year. If 1,000 vehicles will be sold next month, what is the best estimate (based on last year's sales) of the number of two-door cars that will be sold?

59. In a scale drawing of a rectangular garden, the length is 15 inches and the width is 9 inches. In the drawing, 2 inches represents 3 yards. What is the width of the actual garden, in yards?

60. A kindergarten teacher has 72 crayons, 60 pencils, and 84 sheets of paper to distribute to the children in her class. If each child receives an equal number of each item and there are no items remaining, what is the greatest possible number of children in the class?

-
- 61.** Mr. Smith opened a retirement account with a deposit of \$900. This account earns 5% simple interest annually. How many years will it take for his \$900 deposit to earn \$360 interest?

- 62.** Solve the equation for x :

$$\frac{(21)(14)x}{(49)(48)} = 0.875$$

MULTIPLE CHOICE QUESTIONS

QUESTIONS 63–114

DIRECTIONS: Solve each problem. Select the answer from the choices given. Mark the letter of your answer on the answer sheet. When you are solving problems, you can write in the test booklet or on the scrap paper given to you.

- 63.** What is the value of

$$3\frac{1}{4} - 1\frac{2}{3} - \left(-4\frac{1}{2}\right)?$$

A. $-3\frac{1}{12}$

B. $-2\frac{11}{12}$

C. $6\frac{1}{12}$

D. $9\frac{5}{12}$

-
- 64.** On Fran’s map, 2 centimeters represents 3 kilometers. The points on this map that represent Fran’s home and Bryant State Park are 4.5 centimeters apart. How many kilometers apart are Fran’s home and Bryant State Park, to the nearest kilometer?

E. 3

F. 5

G. 7

H. 9

- 65.** Solve for x in terms of y if $2x + 4 = y$.

A. $x = \frac{y - 4}{2}$

B. $x = 2(y + 4)$

C. $x = \frac{y + 4}{2}$

D. $x = 2(4 - y)$

-
- 66.** Bryan completed a 100-meter race in 11.74 seconds. Luis completed the same race in 11.69 seconds. What was the difference between their times, expressed as a fraction of a second?

E. $\frac{1}{20}$

F. $\frac{1}{10}$

G. $\frac{3}{20}$

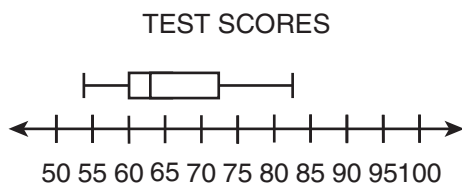
H. $\frac{1}{5}$

- 67.** A total of 24 students in a math class took a 100-point test:

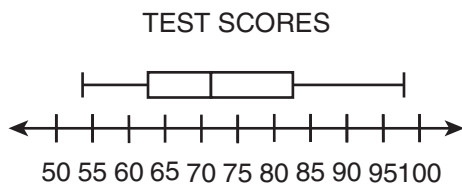
- None of the students scored 100 points.
- 75% of the students scored less than 82 points.
- The lowest score in the class was 54 points.

Which box plot best represents the test scores from this math class?

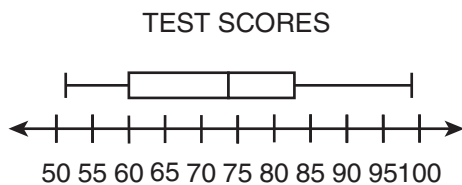
A.



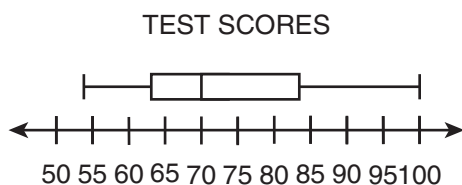
B.



C.



D.



68.

$$n = 0.13 + \frac{207}{100}$$

What is the value of n expressed as a fraction?

E. $\frac{1}{3}$

F. $\frac{11}{5}$

G. $\frac{9}{4}$

H. $\frac{23}{10}$

- 69.** The perimeter of a 12-sided polygon is 100 centimeters. The length of one side is 25 centimeters. The length of another side is 35 centimeters. The remaining sides are equal in length to one another. What is the length of each of the remaining sides, in centimeters?

A. 4

B. 5

C. 20

D. 40

- 70.** Shioh-Chen has a box that contains 4 green cards, 7 yellow cards, and 6 red cards. She randomly draws one card at a time from the box and does not return the cards to the box after they have been drawn. The first card she draws is yellow. The second card she draws is green. What is the probability that the third card she draws will also be green?

E. $\frac{1}{5}$

F. $\frac{4}{17}$

G. $\frac{1}{4}$

H. $\frac{1}{3}$

- 71.** Each number in a sequence is 3 more than twice the number that comes just before it. If 93 is a number in the sequence, what number comes just before it?

- A.** 92
- B.** 90
- C.** 48
- D.** 45

- 72.** On a map, the distance between Deshawn's house and Xin's house is 5.4 inches. If 2 inches on the map represents 3 miles, what is the distance between these two houses, in miles?

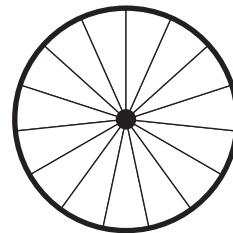
E. 2.6

F. 3.6

G. 7.5

H. 8.1

73.



This wheel has 15 spokes, equally spaced. What is the measure of the angle between adjacent spokes?

- A.** 30°
- B.** 24°
- C.** 15°
- D.** 12°

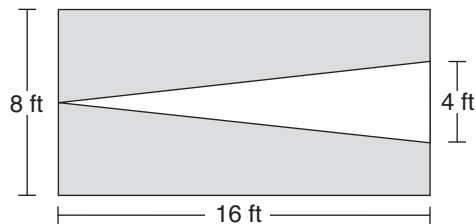
74.

$$m \cdot t = 72$$

For the equation above, what is the sum of the values of m when $t = 1$, when $t = 2$, and when $t = 3$?

- E. 12
- F. 72
- G. 132
- H. 135

75.



The diagram above shows a rectangle with an inscribed triangle. The triangle's vertex and base touch the sides of the rectangle as shown. What is the total area of the shaded portion?

- A. 32 sq ft
- B. 64 sq ft
- C. 96 sq ft
- D. 128 sq ft

76. Orange juice costs \$2 per gallon, and grapefruit juice costs \$3 per gallon. A recent shipment of equal amounts of the two types of juice costs a total of \$250. How many gallons of orange juice are in the shipment?

- E. 50
- F. 62.5
- G. 100
- H. 125

77. Five consecutive numbers have a sum of 110. What is the **least** of these consecutive numbers?

- A. 11
- B. 18
- C. 20
- D. 22

78.

$$\begin{array}{c} -mn \\ (-m)(-n) \\ |mn| \\ |-mn| \\ |(-m)(-n)| \end{array}$$

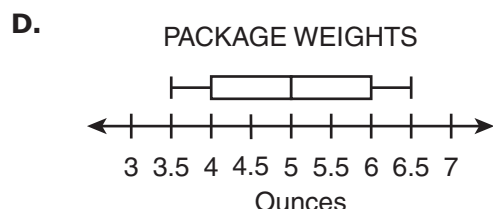
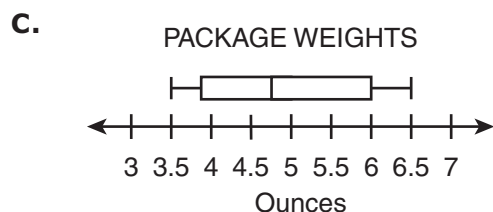
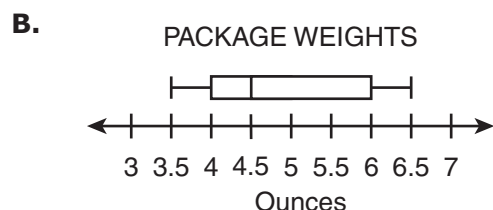
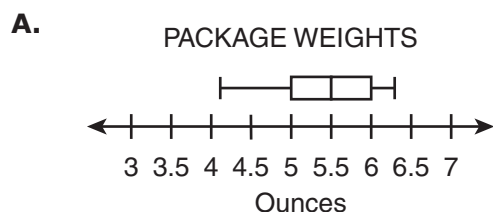
If m and n are positive integers, how many of the five expressions above are equivalent to mn ?

- E. 1
- F. 2
- G. 3
- H. 4

- 79.** A vendor is selling 10 packages of homemade fudge at a bake sale. The vendor weighed and labeled each package. The package weights are given in ounces:

4.6 6.0 3.8 3.5 4.5
5.4 3.9 6.5 4.8 6.2

The vendor created a box plot of the package weights. Which box plot best represents the data?



80.

FRUIT SOLD
AT STORE XYZ

Fruit	Number Sold
Apples	8
Bananas	6
Cherries	12
Oranges	6
Peaches	?
Total	50

Peaches made up what percentage of the total number of pieces of fruit sold at Store XYZ, according to the table above?

- E.** 18%
F. 28%
G. 36%
H. 64%

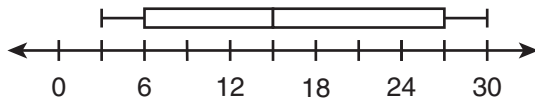
- 81.** A prism is made up of 9 congruent cubes. Each of the cubes has side lengths measuring 4 units. What is the volume, in cubic units, of the prism?

- A.** 108
B. 144
C. 480
D. 576

- 82.** The town of Waterville covers an area of 50 square miles. In 2005, its population was 84,000. In 2006, its population had decreased so that the average population per square mile was equal to 1,500. What was the total decrease in population from 2005 to 2006?

E. 7,500
F. 8,400
G. 9,000
H. 9,200

- 83.** The box plot below represents a set of data.



What is the interquartile range of the set of data?

A. 9
B. 12
C. 21
D. 27

- 84.** $N = \{6, 7, \dots, 21, 22\}$

Set L (not shown) is a list of 25 consecutive integers. The median of the integers in set N above is equal to the least integer in set L. What is the median of the integers in set L?

E. 25
F. 26
G. 27
H. 28

- 85.** The sum of the numbers x , y , and z is 40. The ratio of x to y is 1:2, and the ratio of y to z is 2:5. What is the value of y ?

A. 2
B. 4
C. 8
D. 10

86. Which number could be the sum of 4 consecutive integers?

E. -12
F. -10
G. -8
H. -4

87.

PRICE OF A USED VIDEO GAME AT
VARIOUS STORES

Price of Used Video Game	Number of Stores
\$10.99	5
\$11.99	2
\$12.99	1
\$13.99	4

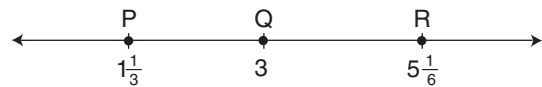
The table above shows the price of a certain used video game at 12 different stores. What is the median price of this video game?

A. \$10.99
B. \$11.49
C. \$11.99
D. \$12.32

88. If $xy = 30$, what is the value of $10 \div x$ in terms of y ?

E. $\frac{1}{10}y$
F. $\frac{1}{3}y$
G. $3y$
H. $10y$

89.



On the number line above, how many units longer is $\overline{QR}$ than $\overline{PQ}$?

A. $\frac{1}{6}$
B. $\frac{1}{2}$
C. $\frac{2}{3}$
D. 3

90.



The distance from W to X is 70% of the distance from W to Y. If the distance from X to Y is 15 miles, what is the distance, in miles, from W to X?

- E. 10.5
- F. 35
- G. 45
- H. 50

91. There are 3 different highways from City W to City X, 4 different highways from City X to City Y, and 3 different highways from City Y to City Z. How many different routes are there for a trip from City W to City X to City Y to City Z?

- A. 10
- B. 12
- C. 24
- D. 36

92. In Center City, the ratio of people to dogs is 4:1, and the ratio of dogs to cats is 3:2. What is the ratio of people to cats?

- E. 2:1
- F. 7:3
- G. 6:1
- H. 12:1

93. The side length of square S is shorter than the side length of square T by 1 centimeter. The area of square S is 9 square centimeters. What is the difference, in square centimeters, between the area of square T and the area of square S?

- A. 3
- B. 7
- C. 8
- D. 10

94. Which expression is equivalent

to $\frac{1}{2}x + 7 - \left(x - \frac{9}{2}\right)$?

E. $\frac{1}{2}(x - 5)$

F. $\frac{1}{2}(x + 5)$

G. $-\frac{1}{2}(x + 23)$

H. $-\frac{1}{2}(x - 23)$

95. How is $\frac{1}{x} - \frac{1}{2x}$ written as a single fraction? (Assume $x \neq 0$.)

A. $\frac{1}{2x}$

B. $\frac{1}{3x}$

C. $-\frac{1}{x}$

D. $-\frac{1}{2x}$

96. In a certain competition, 1 person finished in first place, 5 people tied for second place, 10 people tied for third place, and the other 4,984 people did not place. What percentage of the people in the competition placed first, second, or third altogether?

E. 0.0032%

F. 0.032%

G. 0.32%

H. 3.2%

97. Angle A and angle B are complementary. If the measure of angle B is three times as great as the measure of angle A, what is the measure of angle B?

A. 45°

B. 67.5°

C. 75°

D. 135°

-
- 98.** The height of a tree in 2013 is 300% of the height it was in 2003. By what percentage did its height increase over this ten-year period?

E. $33\frac{1}{3}\%$
F. 100%
G. 200%
H. 300%

-
- 99.** 1 gallon = 16 cups
 1 cup = 8 fluid ounces

If 1 gallon = 2^x fluid ounces, what is the value of x ?

A. 4
B. 7
C. 8
D. 64

- 100.** A certain square puzzle has an area of 64 square inches, and a cubic block has a volume of 64 cubic inches. What is the ratio of the length of a side of the puzzle to the length of a side of the block?

E. 1:3
F. 1:2
G. 2:3
H. 2:1

-
- 101.** In a certain class, Rosa will take a total of 4 tests. She has already taken 2 of the tests and earned scores of 81 and 83. What is the **least** possible score Rosa can earn on the third test and still be able to finish the class with an average score of 85 on all 4 tests? (Assume that test scores can range from 0 to 100.)

A. 76
B. 82
C. 88
D. 90

- 102.** The Barnes family and the Ramirez family each have a pizza of equal size. The first pizza is cut into 18 equal slices, and the second pizza is cut into 15 equal slices. If the Barnes family eats 11 slices from the first pizza, what is the greatest number of whole slices from the second pizza that the Ramirez family can eat without eating a greater percentage of a pizza than the Barnes family ate?

E. 6
F. 7
G. 8
H. 9

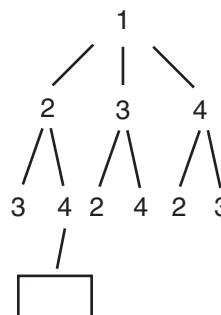
103. $0^3 + 4^0 + 2^{-1} + (-1)^2$

What is the value of the expression above, in decimal form?

A. 0
B. 0.5
C. 1.5
D. 2.5

- 104.** Four slips of paper numbered 1, 2, 3, and 4 were placed in a bag:

- One slip was selected from the bag, and the number recorded.
- Without replacement, a second slip was chosen.
- This process continued until no more slips remained in the bag.
- One part of a tree diagram representing the sample space is shown.



Which number should be placed in the box to complete this part of the tree diagram?

E. 1
F. 2
G. 3
H. 4

105. The outdoor temperature was 4.2°F at 10 p.m. After $3\frac{1}{2}$ hours, the outdoor temperature was -0.7°F . Which value best represents the average change in temperature per hour?

- A.** -1.4°F
- B.** -1.0°F
- C.** 1.0°F
- D.** 1.4°F

106. A rectangular swimming pool of uniform depth is 20 yards long and 14 yards wide. If the water is 6 **feet** deep, what is the volume of water in the pool, in cubic yards?

- E.** 40
- F.** 286
- G.** 560
- H.** 1,680

107. $\{3, 4, 5, 6, 7\}$

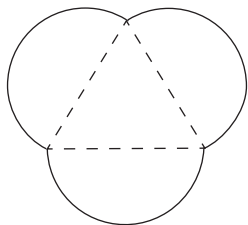
A student made a list of all possible products of 2 **different** numbers in the set above. What fraction of the products are odd?

- A.** $\frac{3}{20}$
- B.** $\frac{3}{10}$
- C.** $\frac{1}{2}$
- D.** $\frac{3}{5}$

108. A number, n , is divided by 25. If the quotient is 10 with a remainder of 5, what is the quotient expressed as a decimal?

- E.** 10.20
- F.** 10.25
- G.** 10.50
- H.** 10.75

109.



The garden design shown above is created in the shape of three semicircles whose diameters lie on the sides of an equilateral triangle. If each diameter is 2 meters long, what is the perimeter of the garden, in meters?

- A. 6
- B. 12
- C. 3π
- D. 6π

110. Micha is playing a game with five cards numbered 1 through 5. He will place the cards in a bag and draw one card at random three times, replacing the card each time. To win a prize, he must draw the number 5 all three times. What is the probability he will draw the number 5 all three times?

- E. $\frac{1}{125}$
- F. $\frac{3}{125}$
- G. $\frac{1}{15}$
- H. $\frac{3}{5}$

- 111.** The speed 10 meters per second is equivalent to how many feet per hour?
(Use the approximation
1 foot = 0.3 meter.)

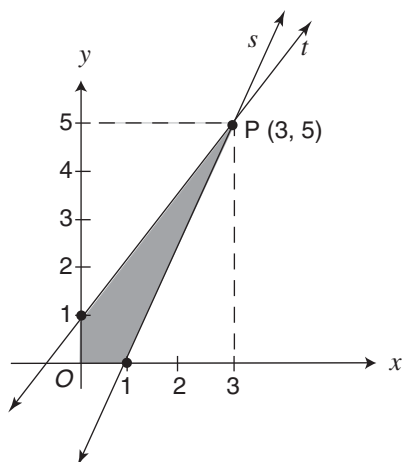
A. 10,800
B. 12,000
C. 36,000
D. 120,000

- 112.** $6.\overline{34} - 4.\overline{12}$

What is the value of the expression above,
written as an improper fraction?

E. $\frac{24}{11}$
F. $\frac{224}{101}$
G. $\frac{111}{50}$
H. $\frac{20}{9}$

113.



Lines s and t intersect at point $P (3, 5)$. What is the area, in square units, of the shaded region in the graph above?

- A. 2
- B. 3
- C. 4
- D. 7

114. A box contains 14 red marbles, 10 black marbles, and 16 white marbles. How many more white marbles need to be added to the box so that the probability of randomly drawing a white marble is $\frac{3}{5}$?

- E. 8
- F. 10
- G. 14
- H. 20

THIS IS THE END OF THE TEST.
IF TIME REMAINS, YOU SHOULD CHECK
YOUR ANSWERS. BE SURE THAT THERE
ARE NO STRAY MARKS, PARTIALLY
FILLED ANSWER CIRCLES, OR
INCOMPLETE ERASURES ON YOUR
ANSWER SHEET. ■